



Dealer Hedging



q-v Mar 26

##

OI (Open Interest)

Open Interest (OI) is the total number of outstanding (open) options contracts that have not been closed, exercised, or expired. It represents real market commitment — each contract is counted once (one buyer + one seller). OI is a cumulative metric updated daily after market close and shows how much "skin in the game" exists at each strike and expiration.

It does **not** reset daily like volume. High OI at a strike often acts as a magnet or barrier because dealers hedge those positions.

OI Put/Call

This is the Put/Call Open Interest Ratio (PCR OI):

$$PCR_{OI} = \frac{\text{Total Put Open Interest}}{\text{Total Call Open Interest}}$$

- 1.0 → more puts open → bearish sentiment or heavy hedging.
- <1.0 → more calls open → bullish sentiment.
- Extreme readings (>1.5 or <0.5) are often contrarian signals (fear = bottom; greed = top).

OI Put/Call is more stable than volume-based PCR and reflects accumulated positioning.

How to work together with DEX & GEX

OI gives the **baseline positioning** (where the big money is stuck). DEX shows the **directional pressure** (net delta flow from that positioning), and GEX shows the **acceleration/hedging sensitivity** (how violently dealers will re-hedge).

- High OI + positive DEX + positive GEX → strong bullish regime with stabilizing flows.
- Rising OI in puts + negative DEX + negative GEX → potential downside acceleration. Traders combine them intraday: OI sets the levels, DEX confirms sentiment, GEX predicts pinning or breakouts. Use OI as the "map", DEX/GEX as the "real-time GPS".

Delta

Delta (Δ) is the **first-order Greek** and the most fundamental measure of directional risk in options. It tells you **how much an option's price is expected to change** for every \$1 move in the underlying asset.

- **Call Delta:** Always positive (0.00 to +1.00). Deep ITM calls approach +1.00 (move almost 1:1 with the stock). OTM calls are near 0.00.
- **Put Delta:** Always negative (−1.00 to 0.00). Deep ITM puts approach −1.00.

Formula (Black-Scholes simplified):

$$\Delta_{\text{call}} = N(d_1), \quad \Delta_{\text{put}} = N(d_1) - 1$$

where $N(d_1)$ is the cumulative normal distribution.

At-the-money (ATM) options usually have $\approx \pm 0.50$ delta. Delta also approximates the **probability that the option finishes in-the-money** (e.g., 0.70 delta call $\approx$ 70% chance of expiring ITM).

Delta is **dynamic** — it changes as the underlying moves, time passes, or volatility shifts (this change is called **Gamma**).

Why it matters: Delta is the **hedging trigger**. Market makers (dealers) stay delta-neutral, so every option trade forces them to buy or sell the underlying to offset delta risk.

Delta Exposure (DEX) Delta Exposure (DEX) is the **aggregate net delta** across the entire options chain, expressed in **share-equivalent terms** (or notional dollars). It shows the **total directional pressure** that dealers must hedge.

How it's calculated (per strike, then summed):

$$\text{EX per strike} = \Delta \times \text{OI} \times \text{Contract Multiplier} \quad (\text{usually 100 for equity/index option})$$

Total DEX = Σ (Call DEX) + Σ (Put DEX). Positive DEX = net bullish pressure (dealers are short delta overall $\rightarrow$ they buy the underlying to hedge). Negative DEX = net bearish pressure.

Dealer perspective: When retail/institutions buy calls or sell puts, dealers take the opposite side and become **short delta** $\rightarrow$ they buy stock/futures to hedge $\rightarrow$ **positive DEX** (bullish flow). The reverse creates negative DEX.

DEX is a **real-time sentiment gauge** of institutional options flow. Large positive DEX readings often precede upside moves because hedging flows support price.

Put/Call DEX

This breaks DEX into its two components:

- **Call DEX** $\rightarrow$ total positive delta from all call open interest.

- **Put DEX** → total negative delta from all put open interest.

Put/Call DEX ratio or simple difference highlights which side dominates. Heavy call DEX = bullish (call buying or put selling). Heavy put DEX = bearish (put buying or call selling).

Traders watch the **split** intraday to see if new flow is call-heavy or put-heavy.

net DEX = Call DEX – Put DEX (or total aggregated delta exposure).

- **Positive net DEX** → overall bullish directional bias (dealers hedging by buying the underlying).
- **Negative net DEX** → overall bearish bias.

It is the **single cleanest number** summarizing dealer directional risk for the day or session. Magnitude matters: +10 million share-equivalent DEX is far more powerful than +1 million.

0dte and 1dte DEX, 0DTE (zero days to expiration) and **1DTE** options have **explosive delta sensitivity** because:

- Gamma peaks sharply near expiration.
- Delta changes extremely fast with even small underlying moves.
- Theta decay is immediate → dealers re-hedge constantly.

0DTE/1DTE DEX is the **most volatile and highest-impact** component of total DEX. It drives intraday pinning, violent reversals, or breakouts on expiration days. Most modern market action (especially in SPX) is dominated by 0DTE flow.

Put Wall, Call Wall, Major Wall

These are the **key strike levels** identified from the options chain (often tied to highest gamma but directly affect DEX):

- **Call Wall** → strike with the **largest call gamma / delta cluster** → acts as **resistance**. Dealers are heavily short calls here; as price approaches, they sell into strength to stay neutral.
- **Put Wall** → strike with the **largest put gamma / delta cluster** → acts as **support**. Dealers buy dips near it.
- **Major Wall** → the absolute strongest wall (highest absolute exposure) on either side. Price tends to “stick” or reverse violently at walls until broken.

Walls are where the **biggest DEX concentration** lives, making them self-reinforcing magnets.

How to work together with GEX & OIOI = the **map** (where positions are concentrated and how big they are).

DEX = the **directional bias** (net vote of all that positioning).

GEX = the **sensitivity / acceleration** (how violently dealers will re-hedge when price moves).

Practical combo rules:

- High OI at a strike + positive DEX + positive GEX → **strong bullish pin/support**.
- Rising put OI + negative DEX + negative GEX → **downside acceleration** likely.
- ODTE DEX + high GEX near a wall → intraday pinning or breakout fuel.

Use OI for **levels**, DEX for **sentiment**, GEX for **regime** (stable vs. volatile). This trio is the core of modern dealer-flow trading.

Cumulative Dex Orderflow Cumulative DEX Orderflow is the **running total** of delta created by options trades throughout the trading day (analogous to Cumulative Volume Delta in futures, but options-specific).

Every option trade has an immediate delta impact on dealers:

- Call buy / put sell → positive delta orderflow (dealers hedge by buying underlying).
- Put buy / call sell → negative delta orderflow.

The cumulative line shows whether **net delta pressure** has been building bullish or bearish all day. Divergences (price up but cumulative DEX rolling over) often signal exhaustion.

Cumulative Call DEX Orderflow

Same as above but **only from call-side trades**. Rising line = net call buying pressure (bullish flow accumulating).

Cumulative Put DEX Orderflow Only from put-side trades. Rising line = net put buying pressure (bearish flow).

Aggregate DEX Aggregate DEX is the **final snapshot** of cumulative DEX orderflow at the end of the day (or current session total). It answers: "How much net delta did the entire options market add or remove today?"

- Strongly positive aggregate DEX → dealers ended up hedging by buying → potential support into tomorrow.
- Strongly negative → hedging sales → potential overhead pressure.

It is the **daily scorecard** of institutional options sentiment and one of the best inputs for overnight bias or next-day levels.

Gamma

Gamma (Γ) is the second-order Greek in options pricing. It measures the rate of change in an option's delta (Δ) for every \$1 move in the underlying asset's price (S). Mathematically:

$$\Gamma = \frac{\partial \Delta}{\partial S} = \frac{\partial^2 C}{\partial S^2}$$

(where C is the call price; the same applies to puts). Gamma is always positive for long calls and long puts (the buyer has positive convexity) and negative for short options. It is highest for at-the-money (ATM) options and increases dramatically as time to expiration approaches zero—short-dated options are “gamma-rich.” Gamma tells you how quickly your directional exposure (delta) will change if the market moves, forcing hedgers to rebalance. Tastytrade +1

Gamma Exposure (GEX)

Gamma Exposure (GEX) is the aggregate dollar-notional gamma position of market makers/dealers across all open options strikes for a given underlying (e.g., SPX, SPY, individual stocks). It quantifies how much dealers must buy or sell the underlying to stay delta-neutral as the price moves. GEX is typically calculated per strike as:

$GEX \approx \text{Gamma} \times \text{Open Interest} \times \text{Contract Multiplier} \times \text{Spot Price (in dollar terms)}$.

It is plotted as a “gamma profile” (histogram or ladder) showing positive or negative exposure by strike. Positive GEX (blue in many platforms) means dealers are net long gamma; negative GEX (red) means they are net short gamma. GEX is not static—it shifts intraday with price, new flow, and volatility. Traders use it to forecast mechanical hedging flows that can support, resist, or accelerate price action. Support.spotgamma +1

Put/Call GEX

Put/Call GEX splits the total gamma profile into its call-side and put-side components. Call GEX reflects gamma from call options (usually dealer short calls → negative gamma contribution in bullish regimes). Put GEX reflects gamma from put options (often dealer short puts when clients buy protection). This split reveals asymmetry: heavy call gamma above spot can create resistance; heavy put gamma below spot can create support. Platforms show separate call and put ladders or a net view. Put/Call GEX helps identify whether the dominant hedging pressure is bullish (call-heavy) or bearish (put-heavy). [Gexstream](#)

Dealer Long / Short Gamma

Dealers (market makers) are assumed to be on the opposite side of client flow and hedge to stay delta-neutral. The regime depends on net GEX:

- **Dealer Long Gamma (Positive Net GEX):** Dealers hedge by *selling into rallies* and *buying dips*. This is counter-cyclical / stabilizing → tighter trading ranges, mean reversion, lower realized volatility. Price tends to oscillate between gamma walls.
- **Dealer Short Gamma (Negative Net GEX):** Dealers hedge by *buying into rallies* and *selling dips*. This is trend-following / destabilizing → amplified moves, higher volatility, potential squeezes or crashes. A small move can snowball because hedging adds fuel in the same direction.

Insiderfinance +1

The flip between these regimes is one of the most watched signals in modern options analysis.

Gamma Squeeze / Pinning

- **Gamma Squeeze:** Occurs in a short-gamma regime when a strong directional move forces dealers to chase price. Example: price rallies → dealers (short calls) must buy more underlying to re-hedge → further rally → more buying. Classic examples include meme-stock squeezes (GME, AMC) or rapid index breakouts. The squeeze intensifies near high-gamma strikes.
- **Gamma Pinning:** Near expiration (especially 0DTE/1DTE), price often “pins” to a high-gamma strike because dealers’ hedging flows stabilize the market around that level. The strike acts like a magnet—small deviations trigger corrective hedging that pulls price back. Pinning is strongest when gamma is concentrated and time decay (theta) is extreme. [Stocks.tradingvolatility](#)

net GEX

Net GEX (or Net Gamma Exposure) is the overall dealer gamma position: typically Call GEX minus Put GEX (or total signed gamma). It is the single most important summary statistic:

- Positive net GEX → long-gamma regime (stabilizing, mean-reverting).
- Negative net GEX → short-gamma regime (volatile, momentum-driven).

Many platforms plot net GEX as a line or bar that flips sign at the “zero gamma” level. Traders watch the magnitude and slope of net GEX for regime strength. A large positive net GEX near spot suggests strong support/resistance; a large negative value warns of potential breakouts or breakdowns.[Gexstream](#)

0dte and 1dte GEX

0DTE (zero days to expiration) and 1DTE options have the highest possible gamma because gamma peaks as time to expiration → 0 (while vega → 0). These options are extremely “gamma-rich” and dominate daily volume on indices like SPX/SPY (often >50% of options volume). Their GEX profile changes rapidly intraday:

- Hedging flows are minute-by-minute or even tick-by-tick.
- Pinning, walls, and momentum effects are exaggerated.
- 0DTE GEX is a primary tool for day-trading: it reveals real-time support/resistance, pinning risk, and acceleration zones.
- 1DTE adds overnight carry but still behaves like ultra-short gamma.

Traders separate 0DTE/1DTE profiles from weekly/monthly GEX because the short-dated layer often overrides longer-term positioning intraday.

Put Wall, Call Wall, Major Wall

These are the highest-concentration gamma strikes:

- **Call Wall** = strike with the largest call gamma → primary resistance/ceiling. As price approaches it from below, dealers (short calls) sell underlying aggressively → caps upside or causes reversal.
- **Put Wall** = strike with the largest put gamma → primary support/floor. As price approaches it from above, dealers (short puts) buy underlying → props up price or causes bounce.

- **Major Wall** = the dominant wall (usually the absolute highest gamma strike across calls or puts) or the strongest combined level. It often acts as the "magnet" or "repellent" for the session.

Price frequently oscillates between Put Wall (floor) and Call Wall (ceiling). Breaking a wall can flip the regime and trigger accelerated moves. Optioncharts +1

Zero Gamma / GEX Flip / Vol Trigger

- **Zero Gamma (Gamma Flip):** The exact price level where net GEX crosses zero (positive $\leftrightarrow$ negative). It is the regime-change boundary. Above zero gamma = long-gamma (stable); below = short-gamma (volatile), or vice versa depending on the chart convention—always check the sign.
- **GEX Flip:** The moment price crosses the zero-gamma level, triggering a shift in dealer hedging behavior (stabilizing $\rightarrow$ amplifying or reverse). This is often a high-conviction inflection point.
- **Vol Trigger:** Usually the lowest (or last major) positive-gamma support level. Crossing it downward removes stabilizing gamma and can unleash volatility expansion as dealers flip to short gamma. Many platforms label it explicitly as the "volatility trigger."

Optioncharts +1

(How to combine GEX with DEX & OI)

DEX (Delta Exposure) is the first-order counterpart to GEX: it shows the net directional delta position of dealers (in shares or notional). OI (Open Interest) is the raw number of contracts at each strike—high OI multiplies the impact of gamma.

Practical workflow:

1. High OI + high |GEX| at a strike = "sticky" level (strong wall or pin).
2. Net GEX tells you the regime (stable vs. explosive); DEX tells you the directional bias (dealers net long or short delta overall).
3. Look for confluence: e.g., price near Call Wall + negative net GEX + positive DEX bias $\rightarrow$ strong resistance + potential squeeze if broken.
4. Intraday: monitor GEX flow changes alongside DEX shifts and OI buildup/decline. New OI in ODTE can instantly reshape walls.

5. Rule of thumb: GEX drives acceleration/momentum, DEX drives baseline direction, OI provides magnitude. Use all three together for high-probability support/resistance and regime timing.

Menthora +1

GEX Flow

GEX Flow tracks how gamma exposure changes in real time (intraday updates). It shows:

- Which strikes are gaining/losing gamma as price moves.
- Acceleration or deceleration of hedging pressure.
- New flow entering (e.g., heavy ODTE call buying flipping local GEX).

GEX is dynamic—price movement itself changes the profile. “GEX Flow” dashboards or indicators highlight momentum: rising positive GEX near spot = building support; collapsing GEX = potential break. Especially powerful for ODTE trading: real-time GEX flow lets you anticipate pinning, breakout, or reversal before it happens.[Medium](#)

Net Convexity

Convexity is another name for gamma (second-derivative curvature). Net Convexity is the aggregate net gamma/convexity position of dealers—essentially synonymous with net GEX in most contexts. It quantifies the overall “curvature” the market maker book imparts to price action:

- Positive net convexity → market makers provide mean-reversion (buy low/sell high hedging).
- Negative net convexity → market makers amplify trends (buy high/sell low hedging).

Some platforms plot it as a “convexity ladder” or use it interchangeably with net GEX to describe regime strength. It is the big-picture view of how much “optionality” is embedded in the options market and how that will influence realized volatility and path dependency.

IV (Implied Volatility)

Implied Volatility (IV) is the market's forward-looking estimate of the underlying asset's future price fluctuations, embedded in current option prices. It is solved inversely from pricing models (e.g., Black-Scholes-Merton) so that the theoretical option price matches the observed market price. IV is expressed as an annualized standard deviation (percentage) and is **not** a prediction of direction—only magnitude and uncertainty.

Mathematically, for a call option price C :

$$C = f(S, K, \tau, r, q, \sigma_{IV})$$

where σ_{IV} is the IV that equates the model to market price. Higher IV inflates option premiums (especially out-of-the-money options) because it increases the probability of large moves before expiration. IV is dynamic: it rises with demand for protection (e.g., heavy put buying) and falls when fear subsides. It forms the volatility surface (skew, term structure) and is the key input for pricing new options. Investopedia +1

net IV

Net IV is the open-interest-weighted difference between average call IV and average put IV across all strikes and expirations:

$$\text{Net IV} = \overline{\text{Call IV}}_{\text{OI-weighted}} - \overline{\text{Put IV}}_{\text{OI-weighted}}$$

Positive net IV means calls are relatively more expensive than puts (bullish skew or call-buying dominance). Negative net IV signals puts are richer (bearish protection demand). Platforms compute it by weighting each strike's IV by its open interest, giving more influence to heavily traded levels. Net IV acts as a sentiment gauge: persistent positive net IV often coincides with bullish regimes or low-fear environments; sharp negative spikes flag hedging panics. It is distinct from vega exposure (net vega position) and is purely a call/put IV differential. [Gexstream](#)

RV (Realized Volatility)

Realized Volatility (RV) is the **actual** volatility that has already occurred, measured from historical price data. It is backward-looking and computed as the annualized standard deviation of log returns over a chosen window (daily, intraday, etc.):

$$RV = \sqrt{\frac{252}{N} \sum_{i=1}^N \left(\ln \frac{S_i}{S_{i-1}} \right)^2} \times 100\%$$

(for daily RV; 252 trading days). RV tells you how much the asset actually moved, not what the market priced in advance. It is the benchmark traders compare against IV to decide whether options were “expensive” or “cheap” after the fact. RV spikes during crises and mean-reverts in calm markets. SpotGamma and similar tools display RV alongside IV to highlight mispricings. [Support.spotgamma +1](#)

net RV

Net RV is less standardized than net IV but follows the same logic in advanced platforms: an OI-weighted or differential measure of realized volatility across calls versus puts (or the net realized move implied by the option book). In practice, it often appears as the realized counterpart to net IV—showing whether the actual price action has been more “call-like” or “put-like” in magnitude relative to open interest. Some analytics compute it as the difference in effective realized ranges weighted by strike participation. While net IV reflects **pricing** sentiment, net RV reflects **what actually happened**. It is used to validate whether IV was correctly forecasting the realized path or if the market over-/under-priced volatility in specific regimes. When net RV consistently exceeds net IV expectations, it signals a gamma-scalping-friendly environment. [Pm-research](#)

IV vs RV

This is the core volatility arbitrage relationship:

- **IV > RV** (typical): Options sellers have a statistical edge. Premium collected > actual movement realized → positive theta + gamma scalping profit if RV stays below IV.

- **RV > IV**: Options buyers win. Actual moves exceed what was priced → long-vol strategies (long straddle/strangle, long vega) profit.

The spread (IV – RV) is often called the “volatility risk premium.” Historically, IV overstates RV because investors pay a premium for protection (crash-o-phobia). Traders monitor the IV/RV ratio or IV rank to decide sell-premium vs. buy-vol strategies. In dealer-flow frameworks, when RV starts exceeding IV while net GEX is negative, momentum accelerates because hedging flows chase the real move. Menthorq +2

Vanna

Vanna is a second-order Greek measuring how an option’s delta changes when implied volatility changes (or equivalently, how vega changes with the underlying price):

$$\text{Vanna} = \frac{\partial \Delta}{\partial \sigma} = \frac{\partial \text{Vega}}{\partial S}$$

Positive vanna (common for long calls / short puts) means delta increases as IV rises. Dealers who are short options (standard assumption) therefore have **negative vanna exposure overall** in many regimes. Vanna captures the cross-effect between price and volatility—critical because spot-vol correlation is usually negative (vol spikes when prices fall). Spotgamma +1

Charm

Charm (sometimes called “delta decay” or “third Greek”) measures how delta changes purely with the passage of time, holding all else constant:

$$\text{Charm} = \frac{\partial \Delta}{\partial \tau}$$

(where τ = time to expiration). For out-of-the-money puts that dealers are short, charm is typically positive: as time passes and the put decays, its delta shrinks toward zero, forcing dealers to **buy back** their short delta hedge (supportive flow). Charm is strongest near expiration and for short-dated options. It explains the "overnight grind" or post-OpEx flows. Spotgamma +1

Vanna Exposure

Vanna Exposure (often labeled VEX or Vanna profile) aggregates dealer vanna across the entire options book, strike-by-strike or net. It quantifies how many shares dealers must buy or sell to re-hedge delta **when IV moves 1 point**. Calculation mirrors GEX:

VEX per strike $\approx -(\text{Call OI} \times \text{Vanna}_{\text{call}}) + (\text{Put OI} \times \text{Vanna}_{\text{put}}) \times \text{multipliers}$.

- Positive net Vanna Exposure $\rightarrow$ dealers buy stock when IV rises (or sell when IV falls).
- Negative net Vanna Exposure $\rightarrow$ opposite (classic for short-put books).

Vanna Exposure creates "vol-driven flows" that amplify or dampen price moves when volatility changes. It is especially powerful intraday or around events. Medium +2

Charm Exposure

Charm Exposure (CEX) aggregates the time-decay effect on dealer delta across all strikes. It shows how much delta hedging is required **per unit of time** (often per day or per hour) as options decay. Formula:

CEX per strike $\approx -(\text{Call OI} \times \text{Charm}_{\text{call}}) + (\text{Put OI} \times \text{Charm}_{\text{put}}) \times \text{contract multiplier}$.

- Positive Charm Exposure $\rightarrow$ supportive daily bid (dealers buy back delta as OTM puts decay).
- Negative Charm Exposure $\rightarrow$ daily offer (delta increases, forcing sales).

Charm Exposure is the "conveyor belt" flow—predictable and steady—unlike gamma (price-driven) or vanna (vol-driven). It explains why markets often drift higher in low-vol, positive-gamma regimes. Medium +1

net Vanna/Charm

Net Vanna and Net Charm are the single-summary statistics: total dealer vanna exposure and total dealer charm exposure summed across the entire chain (all strikes, all expirations). They are the volatility and time analogs to **net GEX**:

- **Net Vanna** tells the overall vol-sensitivity of dealer hedging.
- **Net Charm** tells the overall time-decay hedging bias (daily drift).

Platforms plot them as lines or heatmaps. Large positive net charm = structural bid into the close or overnight. Large negative net vanna + rising IV = forced selling pressure. These nets are updated intraday and are key regime indicators alongside net GEX.Unusualwhales +1

Vanna & Charm Flow

Vanna Flow and Charm Flow track **changes** in these exposures intraday or day-over-day. As price, IV, or time moves, the exposure ladder shifts—new strikes gain/lose vanna/charm. "Flow" dashboards highlight:

- Acceleration zones (where vanna/charm is rapidly turning positive/negative).
- Flip points (when net vanna/charm crosses zero).
- Interaction with 0DTE/1DTE layers.

Vanna flow is event-driven (IV shocks, news); charm flow is steady (time passage). Traders watch flow to anticipate **acceleration** or **reversal** of dealer hedging before it hits the tape. Combined with GEX flow, it gives a three-dimensional view of mechanical order flow.Youtube +1

Combining net Vanna/Charm and net IV

This is the full volatility-flow framework:

Metric	What it shows	Interpretation when combined with net IV
Net IV	Call vs Put pricing sentiment	Positive net IV + positive net charm → strong bullish drift (calls rich + time decay supports)
Net Vanna	Vol-driven delta hedging	Negative net IV (put-rich) + negative net vanna → vol spike forces heavy selling (classic crash amplifier)
Net Charm	Time-driven delta hedging	Positive net IV + positive net charm → low-vol grind higher (dealers buying dips steadily)

Practical workflow:

1. Check **net IV** first → overall vol sentiment (bullish/bearish skew).
2. Layer **net Charm** → daily structural bias (supportive or resistive drift).

3. Add **net Vanna** → how sensitive the book is to an IV shock.
4. Cross-reference with **net GEX** for price-level support/resistance.

Confluence (e.g., positive net IV + positive net charm + positive net GEX) = stable, bullish regime with low realized vol. Divergence (negative net IV + negative net vanna + negative net GEX) = high risk of accelerated downside. This combination turns dealer positioning into a predictive edge for intraday swings, overnight gaps, and post-OpEx behavior.

Vega

Vega (v) is the first-order Greek that quantifies how much an option's price changes for every 1 percentage point (1%) move in implied volatility (IV), holding all else constant. It is identical for calls and puts at the same strike and expiration.

Mathematically:

$$v = \frac{\partial C}{\partial \sigma} = S \sqrt{\tau} N'(d_1) e^{-q\tau}$$

(where C is the call price, σ is IV, τ is time to expiration, and $N'(d_1)$ is the standard normal density).

Vega is always positive for long options (buyer benefits from rising IV) and negative for short options. It peaks for at-the-money (ATM) options and increases with longer time to expiration (long-dated options are "vega-rich"). Short-dated options (ODTE/1DTE) have very low vega.

In dealer-flow contexts, market makers are typically short the options they sell to clients → net short vega overall. A 1% IV increase hurts their book (they lose money on the options they are short), while an IV decrease helps them. Vega itself does not trigger direct delta-hedging flows like gamma does; instead, dealers hedge vega risk by trading volatility products (VIX futures, vol swaps, or other options) or by adjusting their overall book.

Vega Exposure

Vega Exposure (sometimes labeled Net Vega or VEG) is the open-interest-weighted aggregate vega position of dealers across the entire options chain (all strikes, all expirations). It is calculated similarly to GEX:

Vega Exposure per strike $\approx$ (Call OI $\times$ vcall) – (Put OI $\times$ vput) $\times$ contract multiplier $\times$ spot scaling

Vega Exposure per strike $\approx$ (Call OI $\times$ vcall) – (Put OI $\times$ vput) $\times$ contract multiplier $\times$ spot scaling

The net figure shows the total dollar change in the dealer book for a 1% parallel shift in IV.

- **Negative net Vega Exposure** (typical regime): Dealers lose money when IV rises $\rightarrow$ they may sell stock (or futures) to offset or hedge via vol products, amplifying downside if a vol spike coincides with a price drop.
- **Positive net Vega Exposure** (rarer, usually after heavy call buying): Dealers benefit from rising IV.

Vega Exposure is the pure “vol sensitivity” of the book. It is distinct from Vanna Exposure (which is the cross-effect of IV changes on delta). Large negative Vega Exposure + rising IV is a classic amplifier of sell-offs, especially when combined with negative net GEX (short-gamma regime).

Theta

Theta (Θ) measures the daily time decay of an option’s extrinsic value, i.e., how much the option price loses as one day passes, holding all else constant. It is the “bleed” that option buyers pay and sellers collect.

Mathematically (for a call):

$$\Theta = -\frac{S\sigma N'(d_1)e^{-q\tau}}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2) + qSe^{-q\tau}N(d_1)$$

Theta is negative for long calls and long puts (value decays) and positive for short options. It accelerates dramatically as expiration approaches (especially 0DTE/1DTE) and is highest for ATM options. Deep ITM or OTM options have lower theta.

In practice, theta is quoted in dollars per day. Dealers, being net short options, are net positive theta — they are “theta positive” and collect decay every day the market stays relatively flat.

4) Theta Exposure

Theta Exposure (sometimes TEX or Net Theta) is the aggregate daily time decay across the entire dealer options book, weighted by open interest. It shows the net dollars dealers expect to collect (or pay) from theta each calendar day, assuming no movement in spot or IV.

Formula mirrors earlier exposures:

$$\text{Net Theta Exposure per strike} \approx -(\text{Call OI} \times \Theta_{\text{call}}) + (\text{Put OI} \times \Theta_{\text{put}}) \times \text{multiplier}$$

- **Positive net Theta Exposure** (standard): Dealers collect premium decay daily → structural tailwind that encourages them to hedge in ways that support the market (buy dips to stay delta-neutral). This is the “carry” that powers slow grinds higher in positive-gamma regimes.
- **Negative net Theta Exposure** (uncommon, usually after massive long-option buying): Dealers are paying decay → they may sell into strength to reduce risk.

Theta Exposure is the steady, predictable “conveyor belt” of dealer P/L, in contrast to gamma (price-driven) or vanna (vol-driven).

5) Theta & Charm

Theta and Charm are intimately linked because both are time-driven Greeks, but they operate on different dimensions:

- **Theta** → direct decay of the option premium (P/L impact).
- **Charm** ($\partial\Delta/\partial\tau$) → how the option’s delta itself changes as time passes (hedging-flow impact).

Together they create the daily “theta-charm cycle”:

- As time decays (theta), OTM puts that dealers are short lose delta (positive charm). Dealers therefore buy back some of their short delta hedge → supportive flow (bid into the close or overnight).
- ATM options have the highest combined theta + charm effect near expiration.
- In positive net Charm + positive Theta Exposure regimes, the market experiences a structural daily bid: dealers are collecting premium (theta) while simultaneously being forced to buy stock as deltas shrink (charm). This is why low-vol, positive-gamma environments often see quiet overnight drifts or “power hour” buying.

Platforms that show both often plot them side-by-side or as a combined "Theta-Charm Flow" to highlight the mechanical daily rebalancing pressure. The interaction explains why pinning is strongest on expiration days (theta explosion + charm concentration).

Rho

Rho (ρ) measures the sensitivity of an option's price to a 1 percentage point change in the risk-free interest rate (r), holding everything else constant.

Mathematically (call):

$$\rho_{\text{call}} = K\tau e^{-r\tau} N(d_2)$$

- Calls have positive rho (higher rates increase call values because of the forward price effect).
- Puts have negative rho.

Rho is small for short-dated options and grows linearly with time to expiration → it matters most for LEAPs or long-term equity anticipation securities. In today's low-rate world, rho was largely ignored until the 2022–2023 rate-hike cycle. A 25 bp rate cut can noticeably lift long-dated call prices and suppress put prices.

Rho Exposure

Rho Exposure is the open-interest-weighted net rho of the entire dealer book. It shows how much the dealer options position would gain or lose if interest rates moved 1% (100 bp) in parallel.

Net Rho Exposure is usually small on a daily basis because rates move slowly, but it becomes relevant around FOMC meetings, major central-bank decisions, or when the yield curve shifts dramatically.

- **Positive net Rho Exposure:** Dealers benefit from rising rates (more long calls in the book).
- **Negative net Rho Exposure** (more common): Dealers are net short calls or long puts → they lose if rates rise (or gain if rates fall).

Because rho hedging is not done via stock (unlike delta/gamma), dealers adjust rho exposure using interest-rate futures, bonds, or by trading longer-dated options. In practice, rho exposure is monitored more for macro regime shifts than for intraday flows. When combined with the vol surface (term structure), a steepening term structure + negative net rho can signal expectations of prolonged higher rates.

Skew (Put/Call)

Skew (also called volatility skew or put/call skew) measures the **asymmetry** in implied volatility (IV) across different strikes **for the same expiration**. In equity and index markets, the typical pattern is **negative skew** (or reverse skew): out-of-the-money (OTM) puts trade at significantly higher IV than equidistant OTM calls. This reflects “crash-o-phobia” — investors pay a premium for downside protection because left-tail events (sharp drops) are feared more than right-tail rallies.

Key ways to quantify skew:

- **25Δ Skew** (most common): IV of the 25-delta put minus IV of the 25-delta call. Positive number = put skew (normal for stocks).
- **Risk Reversal (RR)**: Price of selling an OTM call and buying an OTM put (or vice versa). A positive RR skew means puts are richer.
- **Put/Call Skew Ratio**: Often shown as a single number or ladder on platforms (e.g., SpotGamma, SqueezeMetrics).

Mathematically, for fixed expiration τ and spot S , IV is a function of strike K :

$$\sigma(K) = \sigma_{\text{ATM}} + \text{skew term} \times \left(\frac{K - S}{S} \right)$$

The skew term is negative for equities → higher σ for $K < S$ (puts).

Why skew matters for flows:

- Steep put skew → dealers are short OTM puts (clients buying protection) → positive charm + negative vanna exposure.
- Flattening skew → reduced fear, often bullish signal.
- Skew spikes during sell-offs (puts get bid hard) → can trigger vanna-driven selling as IV rises and dealers re-hedge delta.

In the context of earlier topics, skew directly feeds into **net IV** (call IV vs put IV differential) and **net Vanna**, because vanna is largest where skew is steepest.

Term Structure

Term structure (or volatility term structure / vol term) plots **implied volatility against time to expiration** for at-the-money (ATM) options. It answers: "How does the market expect volatility to evolve over different horizons?"

Typical shapes:

- **Contango (upward sloping)**: Short-term IV < longer-term IV. Normal in calm markets; reflects mean-reversion of vol. Traders sell short-dated vol and buy long-dated.
- **Backwardation (inverted)**: Short-term IV > longer-term IV. Signals acute near-term fear (earnings, elections, FOMC). Common before events or during crises.
- **Humped**: IV peaks at intermediate expirations (e.g., 1–3 months).

Mathematically, for ATM options:

$$\sigma_{\text{ATM}}(\tau) = \text{function of } \tau$$

Platforms show this as a curve (often with 1DTE, 1W, 1M, 3M, 6M, 1Y points). The slope is called the "vol curve" or "calendar spread."

Practical trading implications:

- Steep contango + positive net charm → classic "vol decay" environment where short-dated options bleed value (theta advantage for sellers).
- Backwardation + negative net GEX → explosive short-term moves because high short-dated IV amplifies vanna and gamma flows.

- Term-structure roll-down: As options age from 30DTE to 0DTE, IV can change dramatically if the curve is steep — this is why 0DTE/1DTE GEX behaves differently from longer-dated gamma.

Term structure interacts heavily with **charm exposure** (time-decay flows) and **net IV** because short-dated puts often drive the skew.

Vol Surface

The Volatility Surface (vol surface) is the **complete 3D picture** that combines skew and term structure: implied volatility as a function of **both strike (moneyness)** and **time to expiration**.

Visually:

- X-axis: Moneyness (K/S or delta)
- Y-axis: Time to expiration (τ)
- Z-axis: Implied Volatility σ

It is usually displayed as:

- A 3D mesh/surface plot, or
- Heatmap (color-coded IV by strike $\times$ expiry), or
- Slices (fixed- τ skew curves or fixed-moneyness term curves).

Key features traders watch:

- **Skew slope** at each tenor (how steep the put wing is).
- **Term slope** at each moneyness (how the curve bends).
- **Smile / smirk** shape (OTM wings higher than ATM).
- **Dynamics**: How the entire surface moves when spot changes or vol shocks hit (parallel shift, skew steepening, term flattening).

Mathematically, the surface is $\sigma(S, K, \tau)$. Pricing engines interpolate the surface to value any exotic or custom strike/expiry.

Zomma

Zomma (also known as DGammaDVol) is a **third-order Greek** that measures the rate of change in an option's gamma (Γ) for a 1 percentage point change in implied volatility (σ), holding all else constant. It quantifies how the curvature (gamma) of an option's payoff responds to volatility shocks.

Mathematically:

$$\text{Zomma} = \frac{\partial \Gamma}{\partial \sigma} = \frac{\partial^3 C}{\partial S^2 \partial \sigma} = \frac{\partial \text{Vanna}}{\partial S}$$

Zomma is typically largest for at-the-money (ATM) options and can be negative for many positions (gamma tends to decrease as volatility rises, because higher vol "flattens" the delta curve). In dealer books (where market makers are usually short options), a large negative zomma means that a vol spike will *reduce* their gamma exposure, potentially weakening support/resistance levels that gamma previously provided. Zomma captures "vol-of-gamma" risk and becomes critical during volatility events when the entire gamma profile can shift rapidly.

2) Zomma Exposure (ZEX)

Zomma Exposure (ZEX) is the open-interest-weighted aggregate of dealer zomma across the full options chain. It is calculated analogously to GEX or Vega Exposure:

$$\text{ZEX per strike} \approx (\text{Call OI} \times \text{Zomma}_{\text{call}}) - (\text{Put OI} \times \text{Zomma}_{\text{put}}) \times \text{multipliers} \times \text{spot scale}$$

Net ZEX shows the total change in dealer gamma exposure expected from a 1% parallel IV shift.

- **Positive net ZEX:** A vol increase *amplifies* the dealer gamma profile (gamma walls get stronger or new ones appear).
- **Negative net ZEX** (more common in short-gamma regimes): A vol spike *erodes* gamma exposure, weakening pinning/support and potentially accelerating breaks.

ZEX is the "volatility sensitivity of gamma" and acts as a higher-order amplifier to Vanna Exposure. Platforms that surface it (or equivalent third-order metrics) use it to forecast how GEX ladders will reshape during IV shocks.

Color

Color (also known as DGammaDTime, gamma decay, or Colour) is the third-order Greek that measures how an option's gamma changes purely with the passage of one day (or unit of time), holding spot and volatility constant.

Mathematically:

$$\text{Color} = \frac{\partial \Gamma}{\partial \tau} = \frac{\partial^3 C}{\partial S^2 \partial \tau}$$

(where τ is time to expiration). Color is usually positive for long options near expiration because gamma concentrates dramatically as time decays (especially in 0DTE/1DTE). For dealers who are short gamma, positive color exposure means their gamma position *increases* (becomes more negative) as time passes, forcing more aggressive daily hedging. Color explains why gamma effects intensify toward expiration even without spot movement—it is the “acceleration of gamma bleed.”

Color Exposure (CEX)

Color Exposure (CEX) is the aggregated dealer color position across all strikes and expirations. It tells how much the *entire gamma profile* is expected to change per day due solely to time passage.

Formula mirrors prior exposures:

$$\text{CEX per strike} \approx -(\text{Call OI} \times \text{Color}_{\text{call}}) + (\text{Put OI} \times \text{Color}_{\text{put}}) \times \text{multiplier}$$

- **Positive net CEX:** Gamma exposure (GEX) naturally strengthens or concentrates daily → walls become “stickier” over time.
- **Negative net CEX:** Gamma exposure weakens or disperses daily → support/resistance levels erode even if price is flat.

CEX is the time-driven “meta-flow” on top of Charm Exposure. It is especially powerful in short-dated options where color explodes near expiration.

Speed

Speed (also called DGammaDSpot) is the third-order Greek measuring the rate of change of gamma with respect to a \$1 move in the underlying price.

Mathematically:

$$\text{Speed} = \frac{\partial \Gamma}{\partial S} = \frac{\partial^3 C}{\partial S^3}$$

Speed tells how the gamma *ladder* itself tilts or skews as spot moves. Positive speed near ATM means gamma increases on one side of the strike and decreases on the other, creating asymmetry in how walls behave intraday. It is the “convexity of convexity” and explains why gamma profiles are not symmetric—speed captures the third-derivative curvature that makes hedging flows accelerate or decelerate nonlinearly as price travels through strikes.

6) Speed Exposure (SEX)

Speed Exposure (SEX) aggregates dealer speed across the chain and shows how the overall gamma profile will *tilt or reshape* for every \$1 move in the underlying.

Calculation is parallel to GEX:

$$\text{SEX per strike} \approx (\text{Call OI} \times \text{Speed}_{\text{call}}) - (\text{Put OI} \times \text{Speed}_{\text{put}}) \times \text{multiplier}$$

- **Positive net SEX** in certain regimes: Price moves cause gamma to concentrate or migrate in a way that reinforces the current trend (self-reinforcing).
- **Negative net SEX**: Moves cause gamma to flatten or flip, often creating sudden regime changes or “gamma flips” at unexpected levels.

SEX is the intraday “shape-shifter” of the GEX ladder.

Combining ZEX, CEX, SEX

These three third-order exposures form the **dynamic evolution layer** of the gamma profile. While first- and second-order Greeks (delta/gamma, vanna/charm) tell you *current* flows, ZEX/CEX/SEX tell you *how those flows will change* as spot, time, and vol move.

Practical synthesis table (building on net Vanna/Charm + net IV from earlier):

Metric	Driver	What it changes in GEX profile	Regime signal when net positive
ZEX	Volatility	Gamma sensitivity to IV shocks	Vol spike strengthens walls (amplifies stability or squeezes)
CEX	Time	Daily gamma concentration/decay	Gamma walls "harden" over the session (stronger pinning)
SEX	Spot price	Intraday tilt/skew of gamma ladder	Moves cause gamma to migrate supportively (trend reinforcement)

Workflow:

1. Check **net GEX** first (current price structure).
2. Layer **ZEX** → will a vol move reshape support/resistance?
3. Add **CEX** → will time passage make walls stronger or weaker today?
4. Add **SEX** → as price travels, does the gamma ladder tilt in your favor or against you?

Confluence (positive ZEX + positive CEX + positive SEX near a major wall) = extremely sticky, self-reinforcing levels. Divergence (e.g., negative ZEX while IV is rising) = warning that current gamma support is about to evaporate.

ZEX, CEX, SEX and GEX

GEX is the *static snapshot* of dealer gamma at a given moment. ZEX, CEX, and SEX are the *velocity and acceleration* of that snapshot:

- **GEX** = where the walls are *now*.
- **SEX** = how the walls *tilt* as price moves (intraday reshaping).
- **CEX** = how the walls *decay or concentrate* with time (daily evolution).
- **ZEX** = how the walls *expand or collapse* when vol changes (shock sensitivity).

Together they give a four-dimensional view of dealer hedging:

- Positive net GEX + positive SEX + positive CEX + positive ZEX → ultra-stable regime with self-reinforcing pinning and low realized vol.
- Negative net GEX + negative SEX + negative CEX + negative ZEX → explosive, self-amplifying moves (classic gamma-squeeze or crash setup).

In practice, advanced platforms that display these (or equivalent higher-order ladders) let traders anticipate not just *where* price will pin or break, but *how quickly* and *under what conditions* the entire gamma structure will evolve. This is the highest-resolution layer of mechanical order-flow analysis—price, time, volatility, and now the *rate of change of curvature itself*.

Premium

Premium is the market price of an option contract — the total dollar amount a buyer pays (and a seller collects) per contract. It consists of two components:

- **Intrinsic value** = $\max(0, S - K)$ for calls or $\max(0, K - S)$ for puts (immediate exercise value).
- **Extrinsic value** (time value + volatility value) = everything else, driven by time to expiration, implied volatility, interest rates, dividends, and the Greeks.

Mathematically, the premium of a call is simply the Black-Scholes price CC :

$$C = Se^{-q\tau} N(d_1) - Ke^{-r\tau} N(d_2)$$

In dealer-flow analysis, "Premium" is usually viewed in aggregate terms: the total notional premium across all open interest (OI) or traded volume. Dealers are almost always **net short premium** because clients buy options for speculation or hedging; market makers sell them and collect the premium as compensation for bearing gamma, vega, vanna, etc. risk. High aggregate premium levels signal elevated demand for optionality (often high IV environments) and increase the absolute size of all exposures (GEX, VEX, etc.).

Premium is the raw "fuel" that powers the entire dealer positioning framework. Every dollar of premium collected by dealers must eventually be managed through delta, gamma, vanna, and charm hedging flows.

2) net Premium

Net Premium is the open-interest-weighted (or flow-weighted) dollar difference between total call premium and total put premium across the options chain:

$$\text{Net Premium} = (\text{Total Call Premium}) - (\text{Total Put Premium})$$

where Total Call/Put Premium = $\sum (\text{OI} \times \text{Option Premium} \times \text{Contract Multiplier})$ for all strikes and expirations.

- **Positive net Premium:** More dollar capital is committed to calls than puts (bullish premium skew). Dealers are net short more call premium → typically increases negative gamma and negative vanna exposure in bullish regimes.
- **Negative net Premium:** Puts dominate in dollar terms (bearish protection buying). Dealers are net short more put premium → classic positive charm + negative vanna setup that often supports slow grinds or sets up for vanna-driven selling on vol spikes.

Net Premium is the **money-weighted** counterpart to net IV (which is volatility-weighted). While net IV tells you relative richness of calls vs puts on a percentage basis, net Premium shows the actual dollar imbalance. A large positive net Premium with flat or rising IV is a stronger bullish signal than net IV alone because it reflects real capital commitment. Platforms compute it live and often display it as a line chart or heat-map alongside net GEX to reveal whether premium positioning aligns with gamma positioning.

DEX Ladder

The **DEX Ladder** (Delta Exposure Ladder) is the strike-by-strike visualization of the aggregate **first-order delta** position of dealers/market makers across the entire options chain. It shows, for every strike, how many shares (or futures contracts) dealers must buy or sell to remain delta-neutral if the underlying moves to that exact level.

Calculation per strike (simplified):

$$\text{DEX}_{\text{strike}} = [(\text{Call OI} \times \Delta_{\text{call}}) - (\text{Put OI} \times \Delta_{\text{put}})] \times \text{Contract Multiplier} \times \text{Spot Scaling}$$

The ladder is typically plotted as a histogram or stepped chart:

- **Positive DEX bars** (often green/blue): Dealers are net long delta at that strike → they would sell stock if price rises to it (resistance).
- **Negative DEX bars** (often red): Dealers are net short delta → they would buy stock if price falls to it (support).
- The **zero line** is the current spot price; the cumulative/net DEX is shown as a line or summary number.

Key insights from the DEX Ladder:

- It reveals the **directional bias** of the dealer book that GEX (second-order) does not show directly.
- Large positive DEX clusters above spot act as **overhead supply**; large negative DEX clusters below spot act as **bid support**.
- Intraday, the ladder shifts as price moves through strikes (because delta itself changes via gamma).
- Unlike GEX (which is about acceleration), DEX is about the **instantaneous hedge quantity** dealers hold right now.

Traders use the DEX Ladder in tandem with net GEX: a positive net GEX regime with heavy negative DEX below spot creates strong “buy-the-dip” mechanics because dealers are already short delta and will buy even more on any further drop.

GEX Ladder

The **GEX Ladder** (Gamma Exposure Ladder) is the most famous and widely watched visualization in dealer-flow analysis. It displays the **second-order gamma exposure** strike-by-strike, exactly as introduced in the very first set of topics.

Calculation per strike:

$$\text{GEX}_{\text{strike}} = \Gamma \times \text{OI} \times \text{Contract Multiplier} \times \text{Spot Price (dollar notional)}$$

Visual format:

- Histogram or “ladder” with strikes on the vertical axis.
- Positive GEX (blue/green bars) = dealer long gamma → stabilizing flows.
- Negative GEX (red bars) = dealer short gamma → momentum-amplifying flows.
- **Call Wall** = tallest positive or negative call-side bar (resistance).

- **Put Wall** = tallest put-side bar (support).
- **Major Wall / Zero Gamma** = the dominant level or flip point.

The ladder updates in real time (especially the 0DTE/1DTE slice). As price moves, the entire profile “slides” because gamma peaks at the new ATM strike. The GEX Ladder is the **price-structure backbone** of every other ladder — it tells you where mechanical hedging will pin, squeeze, or accelerate price.

Vanna Ladder

The **Vanna Ladder** (Vanna Exposure Ladder) shows the **cross-sensitivity of delta to implied volatility** on a strike-by-strike basis. It is the visual representation of Vanna Exposure covered earlier.

Calculation per strike:

$$VEX_{\text{strike}} = \frac{\partial \Delta}{\partial \sigma} \times OI \times \text{multipliers}$$

Typical display:

- Separate call and put ladders or a combined net view.
- Color intensity shows magnitude.
- Positive vanna bars: delta increases when IV rises (common in long-call or short-put heavy books).
- Negative vanna bars (dominant in most equity/index markets): delta decreases when IV rises → dealers sell stock on vol spikes.

The Vanna Ladder highlights **vol-driven flow zones**:

- Heavy negative vanna concentrated below spot → a vol spike will force dealers to sell more stock (classic downside accelerator).
- Heavy positive vanna above spot → vol spike would force buying (upside accelerator).

It is especially powerful when overlaid on the GEX Ladder: a GEX wall that also sits in a heavy negative vanna zone can “break” violently if IV jumps.

4) Charm Ladder

The **Charm Ladder** (Charm Exposure Ladder) visualizes the **time-driven change in delta** strike-by-strike. It is the granular view of Charm Exposure.

Calculation per strike:

$$\text{CEX}_{\text{strike}} = \frac{\partial \Delta}{\partial \tau} \times \text{OI} \times \text{multipliers}$$

Visuals:

- Bars show daily (or hourly) delta hedging pressure due solely to time passage.
- Positive charm (common for short OTM puts): as time decays, delta shrinks → dealers buy back short delta (supportive daily bid).
- Negative charm: delta grows with time → dealers sell more (daily offer pressure)